CLIENT CASE STUDY / PROJECT BRIEF

Insurance Analytics & Risk Intelligence Dashboard

Business requirement document shared by the client with the BI / Data Analyst

Client	NorthStar Insurance Group
Headquarters	Mumbai, Maharashtra, India
Industry	General & Health Insurance
Prepared by	Head of Operations
Project type	Business Intelligence: Interactive Dashboard (Power BI)
Data period	January 2023 – December 2026
Expected delivery	7 working days from data handover
Reporting users	Management, Sales, Claims, Underwriting

Note: NorthStar Insurance Group is a sample client created for a portfolio project. The data is a sample dataset.

1. About the Company

NorthStar Insurance Group is a multi-region insurer headquartered in Mumbai, Maharashtra, India, selling Health, Motor, Liability, Marine, Life, Property, Cyber and Travel insurance across Europe, North America, Asia Pacific, Middle East & Africa and Latin America. Policies are sold through five channels: Broker, Agent Network, Online, Bancassurance and Direct. The company manages about 200,000 policies and 47,000 customers.

2. Business Problem

Policy, claims and customer data are stored in separate Excel files maintained by different teams. Reports are prepared by hand every month and the numbers often do not match between teams. Because of this:

- Management cannot see premium, claims and profit together.
- Sales cannot easily compare channels, products and regions.
- Many policies lapse or are cancelled, but there is no clear tracking of renewals.
- The claims team cannot quickly find which lines, regions and claim types create losses.
- Underwriting cannot compare customer profitability with risk category.
- Monthly reporting takes several days before every review meeting.

3. Project Objective

Build one interactive Power BI dashboard that gives all teams a single, reliable view of premium performance, policy retention, claims and loss ratio, and customer risk, with filters so that each team can find answers without asking for a new report.

4. Data Provided

Data will be shared as Excel / CSV files or a SQL Server database. Approximate volumes below.

Table	Records	Main columns
Policies	~200,000	Policy No, Customer ID, Product, Business Line, Region, Channel, Start Date, End Date, Policy Status, New / Renewal, Net Premium, Gross Premium, Commission
Claims	~32,000	Claim ID, Policy No, Claim Type, Claim Severity, Claim Status, Claim Amount, Amount Paid, Incurred Amount, Fraud Flag

Table	Records	Main columns
Customers	~47,000	Customer ID, Customer Name, Region
Risk	Per policy	Risk Score, Risk Category (Low / Medium / High)

Policy Status values: Active, Lapsed, Cancelled, Renewed. **Claim Severity values:** Low, Medium, High, Catastrophic.

5. Business Questions to Answer

Team	Questions
Management	How is the overall portfolio performing? How do premium and profit change month by month?
Sales	Which business lines, products, channels and regions bring the most premium? How do new and renewal policies compare?
Claims	How many claims do we have and how much do we pay? Where are the high-severity claims? Which claim types are most common?
Underwriting	Which customers bring the most premium? Which are profitable and which are not? How is risk spread across the portfolio?

6. KPIs Required

KPI	Definition
Total Policies / Customers	Count of policies / distinct customers
Total Premium	Sum of net premium
Gross Premium	Sum of gross premium
Average Premium	Total Premium / number of policies
Total Commission	Sum of commission paid
Total Claims	Count of claims
Claim Amount / Total Paid	Sum of claimed amount / sum of amount paid
Total Incurred	Sum of incurred claim amount
Loss Ratio	Total Incurred / Total Premium
Estimated Profit	Premium minus incurred claims (and commission, if applicable). Final formula to be confirmed with Finance
Fraud Claims	Count of claims with fraud flag
Renewal Rate	Renewed policies / total policies

7. Dashboard Requirements

The dashboard should have four pages. Each page starts with a row of KPI cards.

Page	KPI cards	Visuals required
1. Executive Overview	Total Policies, Total Customers, Total Premium, Total Claims, Estimated Profit	Monthly premium trend; policy status distribution; premium by line of business; premium by region
2. Policy & Premium	Total Policies, Total Premium, Gross Premium, Average Premium, Total Commission	Top 10 products by premium; premium by channel; new vs. renewal; average premium by product; premium vs. gross premium; policy status by business line (table)

Page	KPI cards	Visuals required
3. Claims & Risk	Total Claims, Claim Amount, Total Paid, Total Incurred, Loss Ratio, Fraud Claims	Claim amount by region; claim severity; premium vs. incurred trend; average risk score by business line; claims by line of business; risk category; claim type
4. Customer & Policy Details	Year, risk category and policy status selectors	Top 10 customers by premium; customer table (premium, claims, incurred, estimated profit, risk); customer profitability chart; policy detail table

8. Filters (Slicers)

Year, Region, Business Line, Product, Channel, Policy Status, New / Renewal, Risk Category, Claim Status and Claim Severity, where relevant to the page. Filters must update all visuals and KPIs on the page.

9. Design Guidelines

- Clean, professional look using the company colours (green and orange).
- KPI cards at the top of every page, same position and size on all pages.
- Easy-to-read charts with data labels; no unnecessary decoration.
- Page navigation tabs at the bottom.
- Works on a standard laptop screen (16:9).

10. Deliverables

- Power BI file (.pbix) with data model and DAX measures
- Four-page interactive dashboard as described above
- PDF export of all pages
- Short document listing KPI definitions and data assumptions
- One handover session (screen share) for the teams

11. Timeline

Day	Milestone
1	Kick-off call and data handover; confirm KPI definitions
2–3	Data check, cleaning and data model; DAX measures
4–5	Dashboard build (all four pages)
6	Client review and feedback; corrections
7	Final delivery and handover

12. Acceptance Criteria

- KPI totals on the dashboard match the source data.
- All filters work and update every visual on the page.
- No blank or error visuals; dashboard loads without long delay.
- All four pages and deliverables are handed over within the agreed timeline.
- Up to two rounds of changes are included.

13. Assumptions & Notes

- Customer data may contain personal names; it must be handled confidentially.
- Currency for all amounts will be confirmed at kick-off.
- Any change in scope (extra pages, new data sources) will be discussed and agreed separately.

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